Resonate at NADI 2026 Shared Task: Robust Speech Recognition and Dialect Identification under Acoustic and Domain Shift

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Abstract

This paper describes the Resonate team's submission to the NADI 2026 shared task on multidialectal Arabic speech processing.[1] We participated in the robust dialectal Arabic ASR (Subtask 1.1) and spoken Arabic dialect identification (Task 2) subtasks, adding no external labelled speech. Our best single ASR model is a fully fine-tuned Cohere Transcribe Arabic, a 2.07B-parameter Arabic speech model, trained on 239.6K rows derived entirely from the 12.8K official clips through voice conversion, in-corpus noise augmentation and short-clip re-segmentation. Our submitted entry combines nine checkpoints of four sibling runs with a pivot-anchored ROVER vote and a per-dialect orthographic canonicaliser, obtaining 50.29% WER and 24.43% CER on the official test set and ranking second of nine teams. Our dialect identification system is a fine-tuned WavLM Base+ encoder with context-aware multi-head factorized attentive pooling under an angular-margin objective, which obtained 94% accuracy on the validation set and 41% on the out-of-domain test set. The ASR comparisons support augmenting scarce dialectal transcripts with acoustic variation, while ADI highlights the need for evaluation across domains.

1 Introduction

Arabic dialects differ from Modern Standard Arabic (MSA) and from each other lexically, not merely phonetically, so a model trained predominantly on MSA, of which Whisper (Radford et al., 2023) is the canonical example, cannot back off gracefully. Dialects also lack a standardised orthography, so two annotators can transcribe the same utterance differently and both be right, and the available recordings are in-the-wild: noisy, band-limited, and spoken by few speakers per dialect.

The NADI shared task series (Abdul-Mageed et al., 2024) addressed text-based dialect identification through its first five iterations before shifting to speech processing in 2025 (Talafha et al., 2025). NADI 2026 (Sullivan et al., 2026) continues in this direction through subtasks that isolate two different failure modes:

- Robust dialectal Arabic ASR (Subtask 1.1), which is data-poor, providing an advertised 13.6 hours of training audio across eight countries.
- Spoken Arabic dialect identification (Task 2), which has ample in-domain data but a blind test set drawn from a *different domain* than the training and development data.

We investigate acoustic variation through voice conversion (Baas et al., 2023) and augmentation, a strategy also used by a dialect identification entrant at NADI 2025 (Talafha et al., 2025). For ASR, we fine-tune a large Arabic speech model using only the official training clips and their derivatives, then combine nine checkpoints with ROVER (Fiscus, 1997) and dialect-specific orthographic canonicalisation. Our strongest single model narrowly outperformed the ensemble on validation but not on the blind test. For ADI, we fine-tune WavLM Base+ with attentive pooling under an angular-margin objective. We refer to the systems as RESONATE-ASR and RESONATE-ADI.

2 Multidialectal Arabic ASR

This subtask covers noisy spoken Arabic across eight country-level dialects.

[1]Code and reproducibility resources for both tasks are available at https://github.com/ResonateResearch/resonate-nadi-2026.

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2.1 Dataset

The NADI 2026 ASR dataset covers Algerian (ALG), Egyptian (EGY), Jordanian (JOR), Mauritanian (MAU), Moroccan (MOR), Palestinian (PAL), Emirati (UAE) and Yemeni (YEM) speech. It provides 12,800 training utterances (1,600 per dialect), 10,808 validation utterances and 4,000 blind-test utterances (500 per dialect), sampled at 16 kHz. Training duration is advertised as 13.6 hours; summing file-header durations gives 54,976.6 s (15.27 hours, rounded to 15.3), rather than a duration measured after augmentation or resampling. We distinguish the *official scorer* for blind-test results, the *local approximation* for full-validation comparisons, and the *training-harness scorer* for the recipe comparisons in Appendix A. Appendix F specifies their differences and calibration.

Validation is unbalanced: ALG has 726 utterances, PAL 900 and YEM 1,183, against 1,599–1,600 for the others. Macro WER is the unweighted mean of the eight per-dialect WERs; pooled WER divides total word errors by total reference words across dialects. Table 1 reports macro averages, with pooled results in its caption.

2.2 Training Data

We use no external labelled speech. All 239,615 rows used to train our strongest single model derive from the 12,800 official training clips (Appendix A, Table 3), with complete dialect coverage (28,548–30,761 rows per dialect). In the final recipe, rather than MUSAN (Snyder et al., 2015) or external impulse responses (Ko et al., 2017), we extracted 62,965 speech-free segments from pauses and residual regions of the competition audio itself and generated room impulse responses procedurally. Denoised copies come from the `MossFormerGAN_SE_16K` enhancer of ClearerVoice-Studio (Zhao et al., 2025) selected by a blind listening pilot (Appendix A).

The largest single stream is voice conversion. We use kNN-VC (Baas et al., 2023), which trains nothing: WavLM-Large layer-6 features of the source are replaced frame-by-frame with their nearest neighbours in a target speaker's pool and resynthesised by a prematched HiFi-GAN vocoder (Kong et al., 2020), preserving transcript and timing exactly. Sources are the neural-denoised copies; targets are up to 10 same-dialect speakers per clip, selected by ReDimNet2 (Yakovlev and Okhotnikov, 2026) embeddings at similarity threshold 0.45 and cluster coherence 0.70.

2.3 Model

We fully fine-tune Cohere Transcribe Arabic (July 2026; Cassini et al., 2026), a 2.066B-parameter model released under the Apache 2.0 license. It comprises a FastConformer encoder and an autoregressive Transformer decoder.[2] The encoder has d_{model}=1280, 48 layers, 8 attention heads, subsampling factor 8 and 128 mel bins. Every parameter is trainable and no adapters are used. Under an identical recipe the multilingual sibling was 10.3 percentage points worse in macro WER (training-harness scorer, 56.51% vs 46.18%), and a pre-existing in-house fine-tune was worse than the untouched base.

Dialect conditioning. The evaluation data were organized by country, so country identity was available from the released task structure and supplied to the decoder as context rather than predicted from the audio. We insert the exact country strings `Algeria`, `Egypt`, `Jordan`, `Mauritania`, `Morocco`, `Palestine`, `UAE` and `Yemen` into the decoder-context slot after the start-of-context token. Context labels are masked with -100 in the training loss. This uses only information supplied with the task input; it does not reveal reference transcripts. Conditioning improves matched validation WER by 1.27 percentage points under the local approximation.

Band matching. Spectral analysis found 99% of corpus energy below $\sim$ 1.7 kHz, with the 3–8 kHz band effectively dead at -55 to -65 dB. Our converted audio carried -41 dB at 3–4 kHz: the vocoder was synthesising energy the evaluation data does not contain. We therefore applied an order-10 Butterworth low-pass filter at 2.5 kHz to the converted pool, reducing WER by 1.04 percentage points under the training-harness scorer (Appendix A). In that run we also restricted SpecAugment (Park et al., 2019) frequency masks to the 80 live mel bins, since 37.5% of the band carries no energy. Appendix D describes a length bias in SpecAugment that we diagnosed but did not deploy.

2.4 System Combination and Post-processing

ROVER combination. Our submitted system is a nine-way pivot-anchored ROVER vote (Fiscus,

[2] Architecture described in the official release: https://huggingface.co/blog/CohereLabs/cohere-transcribe-arabic-07-2026-release.

1997). Each system is Levenshtein-aligned to the pivot's hypothesis and a weighted per-slot vote is taken. Deletions vote for the empty word; insertions relative to the pivot are dropped, so the pivot anchors output length; and ties resolve to the pivot. The nine voters are checkpoints of four training runs, weighted from 1.3 down to 0.75 with the strongest run as pivot. Appendix H describes the recorded validation-pool selection procedure. Table 6 gives the full pool and Appendix C the composition of each run.

Orthographic canonicalisation. Dialectal Arabic transcription has no fixed spelling convention, so correct hypotheses are frequently scored as errors. We therefore build a lexicon of majority surface forms from each dialect's own training transcripts. Tokens are first grouped by three orthographic equivalences: *alif maqsura* with *ya*, *ta marbuta* with *ha*, and the hamzated *alif* forms with bare *alif*.[3] Within a group, the most frequent form becomes canonical if it covers at least 60% of the group's occurrences and the group appears at least 30 times. The per-dialect restriction is essential: a global lexicon *increases* WER by 0.05 percentage points, since orthographic bias is annotator- and dialect-specific and pooling cancels opposite-signed effects.

2.5 Experiments and Results

Training configuration. We fine-tune for two epochs using AdamW (β_2=0.98, weight decay 0.01, gradient clipping 1.0) with layer-wise learning rates of 5×10^{-5}, 2×10^{-5} and 1.25×10^{-5} for the body, encoder and head respectively, a cosine schedule with 3% warmup, label smoothing 0.1 and bf16 precision. Appendix H gives run-specific batch settings. Our strongest single model ran 9,984 steps in 3h32m. Checkpoints are saved every 250 steps and ranked by the mean of the eight per-dialect WERs on a *duration-matched* 512-utterance probe. We decode with beam 10 and `no_repeat_ngram_size` 6.

Official submission. Table 1 reports per-dialect WER and CER for our submitted system, which ranked second of nine teams at 50.29% macro WER and 24.43% macro CER. The top system achieved 43.23% WER and third place 50.41%.

Performance varied across dialects: Levantine and Gulf (JOR, PAL, UAE) reached 33–37% blind-test WER, while Maghrebi (ALG, MOR) and MAU remained hardest, with MAU at 80.97%. The aggregate validation-to-test increase is uneven: EGY's 12.64-percentage-point increase accounts for 57% of the 2.77-percentage-point macro increase. This is a descriptive decomposition, not evidence of a particular causal mechanism.

Single model vs. system combination. Compared like for like, our best *single* model was marginally *ahead* of the submitted ensemble on validation (47.50% for canonicalised E17 versus 47.52%), yet scored 51% on the blind test against 50% (rounded to the precision available for the single model). Validation thus ranked the two systems in the opposite order to the blind test; differences this small on validation do not establish a robust test-set ordering.

Effect of post-processing. On E14, canonicalisation reduces WER by 0.234 percentage points (bootstrap 95% CI: 0.166–0.299); on the submitted ROVER pool it reduces WER from 47.773% to 47.522%, a 0.251-percentage-point improvement. Appendix I gives recorded Arabic examples of the raw, combined and canonicalised outputs.

Effect of the data recipe. Appendix A reports the full ladder. Voice conversion is the strongest single data stream: at matched decoding, 124K converted clips beat the 12.8K real clips by roughly one percentage point, and augmentation on top improves accuracy further.

3 Spoken Arabic Dialect Identification

The dialect identification subtask classifies utterances into 20 Arabic varieties: the 19 country-level dialects of Algeria, Bahrain, Egypt, Iraq, Jordan, Saudi Arabia, Kuwait, Lebanon, Libya, Mauritania, Morocco, Oman, Palestine, Qatar, Sudan, Syria, Tunisia, the UAE and Yemen, plus MSA.

3.1 Datasets

Training and evaluation data. We use the organiser-provided ADI-17 (Shon et al., 2020) and ADI-20 extension (Elleuch et al., 2025), distributed as `ArabicSpeech/ADI17` and `ArabicSpeech/ADI20`; we add no external labelled speech. Starting from the ADI-20-53h subset (up to 53 hours per variety), we draw additional material from these resources' training splits up to a target of 100 hours per variety, and fill remaining shortfalls with voice-converted copies.

[3] In Buckwalter notation, Y→y, p→h, and >, < and | → A.

Split	Metric	ALG	EGY	JOR	MAU	MOR	PAL	UAE	YEM	Macro avg.
Validation	WER	60.91	31.77	36.11	76.14	49.72	33.69	36.55	55.29	**47.52**
	CER	26.84	14.73	14.35	39.94	22.13	13.40	13.74	22.43	**20.94**
Blind test	WER	67.14	44.41	33.66	80.97	50.07	33.39	37.44	55.24	**50.29**
	CER	33.06	27.62	13.33	44.12	22.64	14.09	14.70	25.84	**24.43**

Table 1: WER and CER (%) of the submitted RESONATE-ASR system (ROVER-9 + canonicaliser). Validation uses the local approximation; blind test uses the official scorer. Pooled WER/CER are 46.28/20.53% on validation and 49.44/23.89% on the blind test. Macro averages weight varieties equally; pooled rates weight reference words/characters.

The retained ADI-20 additions supply Bahraini, Tunisian and MSA speech and supplement Jordanian and Sudanese speech. Voice-activity detection was run over all training audio, and clips containing less than 50% speech were dropped *from training only*. Appendix E accounts for the retained training segments by variety and source. Development uses the official ADI validation split of 10,806 utterances, distinct from the 10,808-utterance ASR validation split. The blind test set contains 878 unlabelled utterances from a different domain.

Fixed-window segmentation. Sampling one random crop per clip introduces a subtle imbalance: a dialect's share of gradient updates tracks its *clip count* rather than its total duration, and mean clip length varies fivefold across dialects. We instead tile every clip end-to-end with a 5 s window at 5 s hop, retaining shorter clips and trailing remainders only when they are at least 2 s long. This increases both the number of training segments and the proportion of each clip that reaches the model. A per-epoch ± 2.5 s start jitter prevents identical crops from recurring across epochs.

3.2 Model Architecture

Raw 16 kHz audio is encoded by WavLM Base+ (Chen et al., 2022), exposing the representation entering the Transformer and the outputs of its 12 blocks; the convolutional feature extractor remains frozen. Context-aware multi-head factorized attentive pooling (CA-MHFA; Peng et al., 2025) learns two separate sets of softmax-normalized weights over these 13 representations to form key and value streams, each projected to 128 dimensions. A temporal convolution over the keys produces attention scores for 64 heads. Its nine-frame kernel ("context 9") spans the current encoder frame and four frames on either side, with zero padding at the boundaries. Each head normalizes its scores over valid time steps and pools the values; concatenated head outputs are projected to a 256-dimensional utterance embedding. The classifier predicts 20 varieties under additive angular-margin softmax (Deng et al., 2019), with margin 0.2 and scale 32. The model has 96.8M parameters, of which 92.6M are trainable. No speaker-verification backbone is transferred; an earlier run that did so scored lower.

3.3 Experiments and Results

Training configuration. Training is performed with SpeechBrain (Ravanelli et al., 2021) using AdamW with weight decay 0.01 and a peak learning rate of 1×10^{-4} with five epochs of linear warmup followed by cosine decay. The encoder unfreezes top-down at three blocks per epoch from epoch 2, and the angular margin warms from 0 to 0.2 over three epochs. To enhance robustness, an augmented copy of each batch (MUSAN noise (Snyder et al., 2015) at 0–15 dB SNR, simulated reverberation from the Room Impulse Response and Noise Database (Ko et al., 2017), speed perturbation at 90/100/110%, and frequency-band and time-chunk dropping) is concatenated with the clean copy. Appendix H provides additional batch, seed and augmentation details.

Inference. Utterances up to 30 s are processed in full; longer recordings are restricted to a single central 30 s crop. The model produces one prediction per recording, without averaging multiple windows. Thus inference can use longer speech spans than the 5 s training segments, while speech outside the central crop of a long recording is not evaluated by the model. The effect of this training/inference duration mismatch is not isolated by the comparisons reported here.

Evaluation. We report classification accuracy and the language recognition evaluation (LRE) average detection cost, C_{avg} (Lee et al., 2022). This secondary metric aggregates miss and false-alarm

Split	Accuracy (%) ↑	C_{avg} ↓
Validation	**94**	**0.05**
Blind test	41	0.17

Table 2: Final RESONATE-ADI results. Both rows come from the same system. The blind test set is drawn from a different domain than the training and validation data.

errors across target/non-target varieties; lower values indicate better performance.

Official submission. Table 2 reports our results. RESONATE-ADI reaches 94% accuracy and C_{avg}=0.05 on the validation set, ranking third of eight teams in the development phase. On the 878-utterance out-of-domain blind test set, our system ranks seventh with 41% accuracy and C_{avg}=0.17.

4 Conclusion

Our ASR system combines augmented training data, ROVER and dialect-specific orthographic canonicalisation, ranking second of nine teams at 50.29% macro WER. The ASR comparisons support the utility of manufactured acoustic variation, but do not isolate every component. ADI achieved 94% validation accuracy and 41% blind-test accuracy; these results motivate evaluation on held-out domains without establishing a causal explanation for the difference.

Limitations

Small ASR validation differences did not reliably predict blind-test ordering: the strongest single model and the submitted ensemble differed by only 0.02 percentage points on validation. Our recipe-development comparisons do not isolate every augmentation component or interaction. ASR decoding and orthographic canonicalisation also depend on known country metadata; performance without that input is not established by the submitted-system results. For ADI, training uses 5 s segments whereas inference can use up to 30 s; the comparisons reported here do not isolate the effect of this duration mismatch. The substantial validation/blind-test accuracy difference indicates limited transfer across these evaluation conditions, but does not by itself identify the acoustic, speaker or linguistic factors responsible.

References

Muhammad Abdul-Mageed, Amr Keleg, AbdelRahim Elmadany, Chiyu Zhang, Injy Hamed, Walid Magdy, Houda Bouamor, and Nizar Habash. 2024. NADI 2024: The fifth nuanced Arabic dialect identification shared task. In *Proceedings of the Second Arabic Natural Language Processing Conference*, pages 709–728, Bangkok, Thailand. Association for Computational Linguistics.

Matthew Baas, Benjamin van Niekerk, and Herman Kamper. 2023. Voice conversion with just nearest neighbors. In *Interspeech 2023*.

Shaun Cassini, Sebastian Vincent, Xiaolu Lu, Julian Mack, Dhruti Joshi, and Pierre Richemond. 2026. cohere-transcribe-arabic-07-2026 (revision 0a8193c).

Sanyuan Chen, Chengyi Wang, Zhengyang Chen, Yu Wu, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki Kanda, Takuya Yoshioka, Xiong Xiao, Jian Wu, Long Zhou, Shuo Ren, Yanmin Qian, Yao Qian, Michael Zeng, Xiangzhan Yu, and Furu Wei. 2022. WavLM: Large-scale self-supervised pre-training for full stack speech processing. *IEEE Journal of Selected Topics in Signal Processing*, 16(6):1505–1518.

Jiankang Deng, Jia Guo, Niannan Xue, and Stefanos Zafeiriou. 2019. ArcFace: Additive angular margin loss for deep face recognition. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

Haroun Elleuch, Salima Mdhaffar, Yannick Estève, and Fethi Bougares. 2025. ADI-20: Arabic dialect identification dataset and models. In *Interspeech 2025*, pages 2775–2779.

Jonathan G. Fiscus. 1997. A post-processing system to yield reduced word error rates: Recognizer output voting error reduction (ROVER). In *IEEE Workshop on Automatic Speech Recognition and Understanding (ASRU)*.

Jonatas Grosman. 2021. Fine-tuned XLSR-53 large model for speech recognition in Arabic.

Tom Ko, Vijayaditya Peddinti, Daniel Povey, Michael L. Seltzer, and Sanjeev Khudanpur. 2017. A study on data augmentation of reverberant speech for robust speech recognition. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*.

Jungil Kong, Jaehyeon Kim, and Jaekyoung Bae. 2020. HiFi-GAN: Generative adversarial networks for efficient and high fidelity speech synthesis. In *Advances in Neural Information Processing Systems*.

Yooyoung Lee, Craig Greenberg, Lisa Mason, and Elliot Singer. 2022. NIST 2022 language recognition evaluation plan.

Daniel S. Park, William Chan, Yu Zhang, Chung-Cheng Chiu, Barret Zoph, Ekin D. Cubuk, and Quoc V. Le. 2019. SpecAugment: A simple data augmentation method for automatic speech recognition. In *Interspeech 2019*.

Junyi Peng, Ladislav Mošner, Lin Zhang, Oldřich Plchot, Themos Stafylakis, Lukáš Burget, and Jan Černocký. 2025. CA-MHFA: A context-aware multi-head factorized attentive pooling for SSL-based speaker verification. In *ICASSP 2025 – IEEE International Conference on Acoustics, Speech and Signal Processing*.

Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2023. Robust speech recognition via large-scale weak supervision. In *International Conference on Machine Learning (ICML)*.

Mirco Ravanelli, Titouan Parcollet, Peter Plantinga, Aku Rouhe, Samuele Cornell, Loren Lugosch, Cem Subakan, Nauman Dawalatabad, Abdelwahab Heba, Jianyuan Zhong, Ju-Chieh Chou, Sung-Lin Yeh, Szu-Wei Fu, Chien-Feng Liao, Elena Rastorgueva, François Grondin, William Aris, Hwidong Na, Yan Gao, et al. 2021. SpeechBrain: A general-purpose speech toolkit. *arXiv preprint arXiv:2106.04624*.

Suwon Shon, Ahmed Ali, Younes Samih, Hamdy Mubarak, and James Glass. 2020. ADI17: A fine-grained Arabic dialect identification dataset. In *ICASSP 2020 – IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 8244–8248.

David Snyder, Guoguo Chen, and Daniel Povey. 2015. MUSAN: A music, speech, and noise corpus. *arXiv preprint arXiv:1510.08484*.

Peter Sullivan, Bashar Talafha, Ahmed Ashraf, Fethi Bougares, Haroun Elleuch, Chiyu Zhang, Abdel-Rahim Elmadany, Youssef Mohamed, Salima Mdhaffar, Yannick Estève, Mohamed Elhoseiny, Hamzah Luqman, Nizar Habash, and Muhammad Abdul-Mageed. 2026. NADI-2026: The second multidialectal Arabic speech processing shared task. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.

Bashar Talafha, Hawau Olamide Toyin, Peter Sullivan, AbdelRahim A. Elmadany, Abdurrahman Juma, Amirbek Djanibekov, Chiyu Zhang, Hamad Alshehhi, Hanan Aldarmaki, Mustafa Jarrar, Nizar Habash, and Muhammad Abdul-Mageed. 2025. NADI 2025: The first multidialectal Arabic speech processing shared task. In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 720–733, Suzhou, China. Association for Computational Linguistics.

Ivan Yakovlev and Anton Okhotnikov. 2026. ReDimNet2: Scaling speaker verification via time-pooled dimension reshaping. *arXiv preprint arXiv:2603.11841*.

Shengkui Zhao, Zexu Pan, and Bin Ma. 2025. ClearerVoice-Studio: Bridging advanced speech processing research and practical deployment. In *Interspeech 2025*, pages 2980–2984.

A Training Pool and Data-Recipe Ablation

Stream	Rows
Real clips (×3)	38,400
Neural-denoised (×3)	38,400
kNN-VC clones, 2.5 kHz low-passed	60,000
In-corpus waveform augmentation	48,998
CTC-aligned short clips (gated)	15,417
Channel / additive / full-chain aug.	38,400
Total	**239,615**

Table 3: Training pool for our strongest single ASR model. Every stream derives from the 12,800 official training clips; no external labelled speech is used.

Recipe	Rows	WER (%)
Isolated-stream comparisons		
Base model, zero-shot	—	60.63
Real clips only (E1)	12.8K	53.23
Neural-denoised only (E2)	12.8K	54.67
Denoised + augmentation (E3)	12.8K	54.53
kNN-VC clones (E4)	124K	52.30
kNN-VC + augmentation (E5)	124K	52.04
Mixed-recipe development		
Balanced real : VC ≈ 1:1 (E7)	235K	47.08
No band matching (E10)	—	45.93
+ band matching (E11)	186K	**44.89**

Table 4: Data-recipe comparisons on the validation split, greedy decoding throughout. These figures use the *training-harness scorer*, which folds alif, hamza and tamarbuta and reads roughly 3.5 percentage points lower relative to the official scorer. They are internally comparable but must not be compared against Table 1. The two "+ augmentation" rows and the balanced 235K row use MUSAN noise and OpenSLR impulse responses; the in-corpus-only recipe begins at E10.

Checkpoints are ranked on a duration-matched probe because an earlier probe whose duration distribution did not match validation (38.8% sub-1s utterances against 10.5%) inverted the checkpoint ranking.

The denoiser was selected by a blind listening pilot: the chosen `MossFormerGAN_SE_16K` model left speech intact, whereas the highest-SNR alternative removed speech along with the noise. Appendix G details the row accounting and augmentation policies; the mixed-recipe rows are development comparisons, not a factorial ablation of all components.

B ASR Experimental Comparisons

Table 5 compares individual models and system combinations. Parenthetical identifiers link each configuration to the ROVER composition in Appendix C.

	System	Val	Test
single	Band-matched (E11)	49.62	—
	Short-clip augmented (E15)	49.81	—
	Dialect-conditioned (E14)	48.35	—
	Conditioned rerun (E16b)	47.99	—
	Strongest single model (E17)	47.71	51
ROVER	ROVER-8	47.73	50.51
	ROVER-9, E16b pivot	**47.52**	**50.29**
	ROVER-10, E17 pivot	47.33	—

Table 5: ASR macro WER (%) comparisons. Individual models use training step 6,000, except E17 (8,750). Single-model validation figures are raw decodes; every ROVER row includes canonicalisation. Validation uses the local approximation; blind test uses the official scorer. E17's test result is available only at integer precision. Dashes indicate no reported test score.

C ROVER Pool Composition

#	Run / checkpoint	Weight
1	E16b ck6000 *(pivot)*	1.30
2	E14 ck6000	1.10
3	E14 ck7000	1.05
4	E14 ck4750	1.00
5	E14 ck4500	1.00
6	E11 ck6000	0.90
7	E11 ck5250	0.85
8	E15 ck6000	0.80
9	E15 ck6750	0.75

Table 6: The nine ROVER voters, ordered strongest first. Ties in the per-slot vote resolve to the pivot.

The nine ROVER voters are checkpoints of four training runs that differ in data composition. E11 uses 185,798 rows of real, denoised, converted and augmented copies, band-matched as described in Section 2. E14 uses identical data plus dialect conditioning. E15 uses 207,179 rows, adding all CTC-aligned short clips. E16b is a re-run on E14's manifest, and serves as the control against which the short-clip streams were compared. The strongest single model, E17, adds gated short clips and three heavy acoustic-augmentation streams to reach 239,615 rows (Table 3) and 47.71% macro WER on validation, but was not part of the submitted ensemble.

D SpecAugment Length Bias

Stock SpecAugment (Park et al., 2019) draws time-mask widths of $U(0, 100)$ frames, up to 1.00 s, and can completely mask a sub-second clip while its transcript remains supervised. We refer to the resulting tendency to emit words from near-empty input as *mask-induced insertion and length-prior bias*. An opt-in cap limiting mask width to a fraction of valid frames eliminated full blanking in simulation while leaving clips longer than 4 s unchanged. The cap was never enabled in the reported systems; this analysis is a diagnosis, not a component of the submitted method.

E ADI Training Data Accounting

Table 7 reports the audio represented by the training segment manifest after speech filtering and fixed-window segmentation. The 1,526,577 segments derive from 895,988 source clips and represent 1,883.89 hours. Counts exclude on-the-fly augmentation and repeated exposure across epochs. ADI-17 and ADI-20 columns distinguish the original corpus from the extension's additional material; they do not divide the initial ADI-20-53h subset from subsequent sampling. Start jitter changes the sampled positions across epochs without increasing the nominal durations reported here.

Variety	ADI-17	ADI-20	VC	Total
Algerian	70.64	0.00	23.20	93.84
Bahraini	0.00	99.31	0.00	99.31
Egyptian	96.30	0.00	0.00	96.30
Iraqi	96.21	0.00	0.00	96.21
Jordanian	12.17	25.35	56.92	94.43
Saudi	96.32	0.00	0.00	96.32
Kuwaiti	69.36	0.00	24.71	94.07
Lebanese	77.61	0.00	16.40	94.00
Libyan	77.19	0.00	16.92	94.11
Mauritanian	96.42	0.00	0.00	96.42
Moroccan	35.29	0.00	58.14	93.43
MSA	0.00	65.91	29.31	95.23
Omani	39.09	0.00	52.30	91.39
Palestinian	83.43	0.00	10.47	93.90
Qatari	51.25	0.00	39.07	90.32
Sudanese	35.23	10.73	44.16	90.12
Syrian	80.73	0.00	14.42	95.16
Tunisian	0.00	95.68	0.00	95.68
Emirati	81.90	0.00	12.95	94.85
Yemeni	41.31	0.00	47.49	88.80
Total	1140.44	296.98	446.47	1883.89

Table 7: Retained ADI training audio in hours, computed by summing segment durations by variety and source. VC denotes voice-converted audio. Totals are calculated before rounding.

F ASR Scoring Protocol

Official scorer. Blind-test numbers are the organisers' returned scores. For the submitted ensemble, the detailed result contains 22,670 word errors over 45,857 reference words, giving 49.44% pooled WER, and 52,761 character errors over 220,805 reference characters, giving 23.89% pooled CER. The official macro WER/CER are 50.29/24.43%. Pooled rates are not averages weighted by utterance count.

Local approximation. The text-normalisation pipeline was calibrated against the confirmed E11 checkpoint-6000 organiser/leaderboard result of 49.7195% macro WER, using its eight per-variety WER and eight CER values for comparison. The selected pipeline removes Arabic diacritics, replaces Arabic/ASCII punctuation with spaces, and collapses whitespace on both references and hypotheses. It does not fold alif/hamza, alif maqsura/ya or ta marbuta/ha. Empty normalised references are excluded; WER is computed over words and CER retains inter-word spaces. Per-variety corpus rates are averaged equally for the macro result. This remains an approximation, not the official scoring implementation. The reported local scores are not the separately residual-corrected estimates used in internal reports.

The calibrated local approximation was used for validation checkpoint and pool comparisons. Leaderboard feedback was also used during development as a coarse signal for prioritizing subsequent experimental directions. The submitted ensemble has local-approximation validation macro WER 47.52% and pooled WER 46.28%.

Training-harness scorer. The earlier training scorer additionally folds alif, hamza and ta marbuta. Its WER is approximately 3.5 percentage points lower in the recorded comparisons, but this is not a constant correction across systems. Table 4 uses this scorer and greedy decoding. Its values should not be compared directly with the local-approximation or official-scorer values.

G ASR Augmentation Accounting

Repetition and voice conversion. In Table 3, real clips ($\times 3$) and neural-denoised clips ($\times 3$) mean three identical manifest rows per waveform. The latter reuse one denoised waveform, not three independent denoisings. These repetitions alter

sampling frequency; stochastic training-time feature masking is separate. The isolated VC stream in Table 4 contains 124,457 examples. The mixed recipe instead samples 7,500 converted examples per variety, 60,000 in total, with seed 20260717, then applies the 2.5 kHz low-pass filter. Thus 124K and 60K describe different experimental pools.

In-corpus waveform augmentation. The mixed-recipe stream samples 6,125 real clips per variety with replacement and applies exactly one of three equiprobable transformations: additive in-corpus noise, channel filtering, or speed perturbation (equal choice of factors 0.9 and 1.1). Outputs are band-matched to 2.5 kHz. Two failed audio reads leave 48,998 generated rows rather than 49,000. This differs from the isolated-stream augmentation in E3/E5, which selects uniformly among six transformations: in-corpus noise, MUSAN noise/music, recorded reverberation, simulated reverberation, channel degradation, and speed perturbation.

Additional acoustic views. The final single-model pool adds 12,800 examples each from additive-only, channel and full-chain views, generated with seed 42. Additive-only augmentation mixes corpus-derived noise at uniformly sampled 0–20 dB SNR. Channel augmentation applies a band-pass with probability 0.9 (lower cutoff 200–400 Hz, upper cutoff 3,000–3,600 Hz), a codec with probability 0.8, and additive noise with probability 0.5 at 10–25 dB SNR. Nominal codec probabilities are 0.5/0.3/0.1/0.1 for μ-law/GSM/Opus/MP3, renormalised over available codecs.

Full-chain augmentation independently samples the following stages in order: static gain (probability 0.9; $U(-6, 6)$ dB), spectral tilt (0.3; $U(-6, 6)$ dB/octave), sinusoidal gain modulation (0.3; depth 2–8 dB, rate 0.1–0.5 Hz), clipping (none/soft/hard probabilities 0.7/0.2/0.1), additive noise (0.8; 0–20 dB SNR), and simulated reverberation (0.6; target RT60 0.15–0.6 s). Soft/hard clipping thresholds are sampled from 0.5–0.9/0.7–0.95, respectively. This chain contains neither speed perturbation nor a codec stage. The noise sampler selects the same variety with probability 0.8 and pause/residual/synthetic-PSD sources with nominal probabilities 0.6/0.3/0.1.

CTC-aligned short clips. An Arabic XLSR-53 CTC model (Grosman, 2021), used as a forced aligner, supplies word timestamps for the original training transcripts; the retained labels are contiguous reference words, not pseudo-transcriptions. The builder enumerates one- to six-word spans of 0.43–1.00 s, rejecting source alignments below the within-variety tenth score percentile. Boundaries use inter-word gap midpoints with 35 ms outward padding limited by neighbouring words, or 50 ms at utterance edges; clips receive a 6 ms fade. The parent waveform is low-passed before slicing. Duration/word-count quotas approximate the short-utterance validation distribution, with at most six spans per source and maximum word-span Jaccard overlap 0.65. This produces 21,381 clips for E15. Requiring internal boundary gaps of at least 60 ms and first/last aligned-word extents of at least 40 ms retains 15,417 clips for the gated stream in the strongest single-model recipe. Appendix C identifies which submitted runs consume which streams.

H Additional Reproducibility Details

ASR. The saved training arguments for E11, E14, E15, E16b and E17 specify seed 42, and their model configurations record Transformers 5.12.0. Table 8 gives input batch sizes, gradient accumulation and completed two-epoch trainer runtimes; these exclude preprocessing and do not measure time to the selected intermediate checkpoint. Effective batch size is the input batch size multiplied by the number of gradient-accumulation steps. The submitted ROVER weights form a manually assigned descending schedule applied to validation-ranked systems and evaluated in validation pool comparisons using macro WER. Table 6 preserves the exact submitted weights, which are not model-estimated confidence probabilities.

Run	Batch	Accum.	Time (s)
E11	16	1	8,509
E14	16	3	9,834
E15	12	2	9,025
E16b	16	3	≈10,070
E17	24	1	≈12,700

Table 8: ASR training configuration and full-run trainer time. Batch denotes input batch size; Accum. denotes gradient-accumulation steps.

ADI. ADI uses seed 420 and input batches of 64. Clean inputs are concatenated with one augmented copy. With augmentation probability 1, one or two of the five listed waveform transforms are selected and chained in shuffled order. Frequency dropping uses one to three notches of width 0.05 in

the implementation's normalised frequency coordinates; chunk dropping removes one to five spans of 1,000–2,000 samples (62.5–125 ms at 16 kHz). Training uses bf16 precision. The environment report records Python 3.10.12, CUDA 12.6 and SpeechBrain revision `e5cb1f65b`. Internal validation uses 5 s crops, distinct from official inference.

I Recorded Arabic ASR Examples

Algeria — nadi-Algeria-13915_2

Reference	لعقوبة لمياة عام ان شاء الله سيدي
Raw (E16b)	عقبال مية عام إن شاء الله
ROVER-9	قبل مية عام إن شاء الله
Canonicalized	قبل مية عام ان شاء الله

Word errors (raw / ROVER / canonicalized): 4 / 4 / 3; reference words: 7.

Hamzated to bare alif in the conditional phrase; lexical errors remain.

Morocco — nadi-Morocco-16578_2

Reference	الله يرضي عليك، الله يرضي عليك.
Raw (E16b)	الله يرضى عليك، الله يرضى لك.
ROVER-9	الله يرضى عليك الله يرضى لك
Canonicalized	الله يرضي عليك الله يرضي لك

Word errors (raw / ROVER / canonicalized): 3 / 3 / 1; reference words: 6.

Alif maqsura to ya twice; the final word mismatch remains.

Mauritania — nadi-Mauritania-69462_2

Reference	ذاك معناه انك ليهي تجيني آن شور الدار
Raw (E16b)	ذاك معناه آن ذاك شور ليهي نسيني آن لشور الدار
ROVER-9	ذاك معناه آن إنك ليهي ميتين آن لشور الدار
Canonicalized	ذاك معناه آن انك ليهي ميتين آن لشور الدار

Word errors (raw / ROVER / canonicalized): 5 / 4 / 3; reference words: 8.

Hamzated to bare alif; substantial lexical mismatches remain.

Jordan — nadi-Jordan-38753_1

Reference	احبسوني اشنقوني اعملوا كل اللي بدكم إياه فيي
Raw (E16b)	أحبسوني، أسندوني، أعملوا كل اللي بدكم إياه في
ROVER-9	احبسوني اسندوني اعملوا كل اللي بدكم اياه في
Canonicalized	احبسوني اسندوني اعملوا كل اللي بدكم إياه في

Word errors (raw / ROVER / canonicalized): 4 / 3 / 2; reference words: 8.

Bare to hamzated alif; a verb substitution and word-form error remain.

UAE — nadi-UAE-43966_1

Reference	إن شاء الله، إن شاء الله يدوه، يوّد
Raw (E16b)	ان شاء الله، ان شاء الله يدوه، يوم
ROVER-9	ان شاء الله ان شاء الله يدوه يوم
Canonicalized	إن شاء الله إن شاء الله يدوه يوم

Word errors (raw / ROVER / canonicalized): 3 / 3 / 1; reference words: 8.

Bare to hamzated alif twice; the final word substitution remains.

Figure 1: Selected validation examples from Maghrebi (Algeria, Morocco), Mauritanian, Levantine (Jordan) and Gulf (UAE) speech. Raw hypotheses are from the submitted pivot, E16b step 6,000. All strings are recorded outputs, including punctuation. Word-error counts use the local approximation, which removes punctuation and diacritics on both sides; canonicalisation changes hypothesis spellings only. These examples illustrate orthographic gains and remaining lexical errors, not their population frequency.